P-C-436b
May 16, 1963

SUPERSEDING
Fed. Spec. P-C-436a
June 23, 1954

FEDERAL SPECIFICATION

CLEANING COMPOUND, ALKALI, BOILING VAT (SOAK) OR HYDROSTEAM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one grade of alkaline cleaning compound for use in the hot soak tank cleaning of ferrous and nonferrous alloy parts. It is also for use in hydrosteam units operating on existing steam supply for steam cleaning (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-A-355—Aluminum Alloy, Plate and Sheet 2024
QQ-M-44—Magnesium Alloy Plate and Sheet (AZ31)
SS-A-706—Asphalt; (for use in) Road and Pavement Constructions
PPP-D-729—Drums; Metal, 55-Gallon (For Shipment of Noncorrosive Material)
PPP-P-704—Pails: Shipping, Steel (1 through 12 gallon)

Federal Standards:

Fed Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-15016—Lubricating Oil; General Purpose.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Regulations of Federal Hazardous Substances Labeling Act.

(Public Law 86-613, 86th Congress, S.1283, approved July 12, 1960)

(Application for copies should be addressed to Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Material.** The raw materials used in manufacture of these compounds shall be intimately assembled and processed so as to produce a granular, free flowing product which will remain uniform in composition and show no evidence of segregation or caking during handling or storage.

3.2 **Composition.** The cleaning compound shall consist of sodium metasilicate, monobasic sodium phosphate, trisodium phosphate, nonionic surfactant, and anionic surfactant in the proportions necessary to meet the requirements of this specification (see 6.4).

3.2.1 *Silicate.* Silicate, calculated as SiO_2, shall be not less than 15.1 percent by weight of the cleaning compound when tested as specified in 4.4.8.1.

3.2.2 *Total phosphate.* Total phosphates, calculated as P_2O_5, shall be between 17.7 and 22.0 percent by weight of the cleaning compound when tested as specified in 4.4.8.2.

3.2.3 *Total alkalinity.* The total alkalinity, calculated as Na_2O, shall not exceed 28.0 percent by weight of the cleaning compound when tested as specified in 4.4.8.3.

3.2.4 *Alcohol soluble matter.* The total alcohol soluble matter shall be between 17.3 and 18.4 percent by weight of the cleaning compound when tested as specified in 4.4.8.4.

3.3 **Performance requirements.**

3.3.1 *Cleaning efficiency.* When tested as specified in 4.4.1.1, a 4.75 percent (4.75 grams of compound in 100-ml. solution) distilled water solution of the cleaning compound shall completely remove mineral oil and asphalt soils from steel test panels, the removal of the mineral oil being indicated by freedom from water-break and "residue-pattern" stains.

3.3.2 *Stability.* A 4.75 percent (4.75 grams of compound in 100-ml. solution) distilled water solution of the cleaning compound, after boiling for 40 hours and testing as specified in 4.4.1.2, shall completely remove mineral oil and asphalt soils from steel test panels, and in the cleaning efficiency requirement of 3.3.1.

3.4 **Corrosiveness.** The cleaning compound shall not attack aluminum. When tested in accordance with 4.4.2.2.1, there shall be no loss in weight of aluminum test panels and a maximum of 0.5 mg. increase in weight. When tested galvanically coupled to magnesium test panels as described in 4.4.2.2.2, aluminum test panels shall lose no weight but gain from 0.4 to 1.0 mg. while the magnesium test panels shall gain between 1.0 and 2.0 mg. in weight. In both tests, aluminum shall remain bright, unstained, and unetched.

3.5 **pH value.** The pH value of a distilled water solution of the compound at a concentration of 4.75 percent (4.75 grams of compound in 100 ml. of solution) shall be not more than 12.2 when tested at 25° C., as specified in 4.4.3.

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3.6 Dust forming properties. When tested as specified in 4.4.4, the dust forming properties shall be such that the dust shall settle within a period of 5 seconds.

3.7 Surface tension. The surface tension of a 0.0317 percent solution (0.0317 grams of compound in 100 ml. of solution) of the compound in distilled water at 25°C. shall be not more than 36 dynes per centimeter when determined as specified in 4.4.5.

3.8 Penetration. When tested as specified in 4.4.6, the average penetration shall be not less than 30.0.

3.9 Segregation. When tested as specified in 4.4.7, the difference of the percentages of an ingredient of the compound (such as silicate) taken from different portions of a container shall not exceed 2 percent of the average percentage of that ingredient in the compound. (The compound shall be subjected to the test specified in 4.4.7 at the discretion of the inspection laboratory when segregation of the ingredients appears to have occurred.)

3.10 Workmanship. The compound shall be manufactured in accordance with the best commercial practice to produce a high quality material which is stable and not subject to change during storage in a sealed container.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 *Lot.* For purposes of sampling, a lot shall consist of all material mixed or prepared at one time in one vessel. Material shall be identified by order of production or batch number.

4.2.2 *Sampling for acceptance tests.* A minimum of two samples, each weighing approximately one pound, shall be taken from each lot.

4.2.3 *Sampling for examination of filled containers.* A random sample of filled containers shall be taken from each lot in accordance with MIL-STD-105 at inspection level 1 and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.2.4 *Examination of filled containers.* Each sample filled container selected in accordance with 4.2.3 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory marking; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under required fill, shall be rejected; and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3 Tests. Testing under this specification shall be for the purpose of acceptance of individual lots.

4.3.1 *Acceptance inspection.* Inspection for the acceptance of individual lots shall consist of examining and testing the acceptance samples for all the requirements specified in section 3 (see 4.4). The Gov-

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ernment reserves the right to make any additional tests deemed necessary to determine compliance of the compound with all the requirements of this specification. If any sample is found to be not in compliance with this specification, the entire lot which it represents, if produced by continuous-run process, shall be rejected; or if the lot is produced in batches, all untested batches in the lot shall be rejected together with the batch from which the sample was taken.

4.4 **Test methods.**

4.4.1 *Performance tests.* Performance tests shall consist of:

a. Cleaning efficiency
b. Stability

4.4.1.1 *Cleaning efficiency test.*

4.4.1.1.1 *Preparation of test panels.* Test panels of SAE 1020, 18-20 gage, cold-rolled steel, 2-1/2 by 2-1/2 inches in size, with a 1/4-inch diameter hole placed 1/8 inch from one corner, shall be prepared. Sharp edges shall be smoothed with No. 1 coarse emery cloth. The panels shall then be degreased in c.p. grade acetone. Polish both faces with No. 1 emery cloth stroking in one direction only. Clean in a hot alkaline solution until free from water break. Rinse in water, then dip in absolute ethyl alcohol, and air dry.

4.4.1.1.2 *Application of soils.* The following soils shall be applied to separate panels:

4.4.1.1.2.1 *Mineral oil.* The soiling material shall be Military Symbol Oil 3065 conforming to MIL-L-15016. The cleaned panels shall be suspended on an "S" hook and dipped into a 400-ml. beaker of the oil at a temperature of 26° ± 1°C. They then shall be removed and allowed to drain at a temperature of 26° ± 1°C. for 30 minutes. The remaining drop of oil in the lower corner of each panel is then removed with absorbent cotton. The amount of mineral oil adhering per panel shall be approximately 0.16 gram.

4.4.1.1.2.2 *Asphalt.* The soiling material shall be petroleum asphalt conforming to SS-A-706, grade 85-100, designation AP-3. The soil, heated to approximately 50°C. (see 6.3) shall be spread evenly over one face of the cleaned test panels using the edge of a steel spatula, leaving unsoiled the triangular area at the corner which includes the 1/4-inch hole. The amount of asphalt on each panel shall be 0.32-0.34 g. determined by trial and error. Allow each panel to stand for 15 minutes before stripping.

4.4.1.1.3 *Preparation of cleaning solution.* In a 2-liter beaker, prepare 1600 ml. of a 4.75 percent (4.75 grams of compound in 100 ml. of solution), distilled water solution of the test compound. Bring solution to a boil and keep at this temperature throughout the test. Maintain solution level throughout the test by addition of distilled water.

4.4.1.1.4 *Cleaning procedure.* Three test panels with each type of soil, prepared as specified in 4.4.1.1.2, shall be cleaned successively in the following order; mineral oil, then asphalt. The soiled test panel shall be immersed in the cleaning solution specified in 4.4.1.1.3 by means of an iron hook, one end of which passes through the 1/4-inch hole and the other end over a glass rod placed across the top of the 2-liter beaker. At the end of 3-minute intervals the test panel shall be moved forward and backward, six times in each direction, and then up and down three times in each direction. This agitiation of the panels is applied in the cleaning of the asphalt test panels but not in the cleaning of the mineral oil panels. The panel shall then be removed from the solution for visual inspection of the extent of soil removal and quickly replaced. The entire operation of agitation of the test panel and visual examination shall require not more than 6 seconds. Mineral oil test panels shall be cleaned for 3 min-

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utes, and asphalt-soiled panels for 21 minutes respectively. At the end of the specified cleaning period, or before, if visual examination indicates completeness of soil removal, the panel shall be removed from the cleaning solution and given two 6-second rinses in distilled water (no agitation); there shall be a 4-second drain between rinses. The rinsing solutions shall consist of 2 one-liter beakers, each containing 800 ml. of distilled water at 23° ± 3°C.

4.4.1.1.5 *Measurement of residual soil.*

4.4.1.1.5.1 *Asphalt.* Rinsed asphalt test panels, after treatment specified in 4.4.1.1.4, shall be dried for 15 minutes in an oven at a temperature of 75° ± 1°C., followed by cooling at room temperature for 30 minutes. They shall then be examined for residual soil. Tests shall be run in triplicate.

4.4.1.1.5.2 *Mineral oil.* Rinsed mineral oil test panels, after treatment as specified in 4.4.1.1.4, shall be examined for presence or absence of waterbreak. They shall then be dried for 15 minutes at a temperature of 50° ± 1°C., followed by cooling at room temperature for 30 minutes. They shall then be examined for presence or absence of residual stains ("residue-pattern" stains), the number and approximate size of which are recorded. Tests shall be run in triplicate.

4.4.1.2 *Stability test.*

4.4.1.2.1 *Preparation of solution.* In a 3-liter beaker, prepare 1600 ml. of a 4.75 percent (4.75 grams of compound in 100 ml. of solution) distilled water solution of the test compound. Boil the solution for 40 hours, replacing the water lost by evaporation as necessary. Transfer the solution to a 2-liter beaker, bring to a boil, and keep the solution at a boil throughout the test, maintaining the solution level by addition of distilled water.

4.4.1.2.2 *Stability test procedure.* Using the solution prepared in 4.4.1.2.1, three test panels for each type of soil prepared and soiled in accordance with 4.4.1.1.1 and 4.4.1.1.2, shall be cleaned successively as specified in 4.4.1.1.4 (a total of six test panels). The residual soil on each panel shall be determined in accordance with 4.4.1.1.5.

4.4.2 *Corrosion tests.*

4.4.2.1 *Preparation of test panels.* Test panels shall be cut 3 inches by 3/4 inch in size from approximately 0.034-inch thick sheet aluminum conforming to QQ-A-355, T3 or T4 temper. Sharp edges shall be smoothed with No. 4/0—150 alundum polishing paper. Polish both faces with the same abrasive, stroking in one direction only.

4.4.2.2 *Cleaning of panels.* Panels shall be cleaned with c. p. acetone, using a swab of absorbent cotton. They shall then be wiped with paper toweling, dipped in absolute ethyl alcohol, and again wiped with paper toweling.

4.4.2.2.1 *Corrosion test procedure.* The clean test panel shall be weighed to the nearest 0.0001 gram and immersed completely in 200 ml. of a boiling 4.75 percent (4.75 grams of compound in 100 ml. of solution) distilled water solution of the compound contained in a 500-ml tall-form beaker. The solution shall be kept at a boil throughout the test, the solution level being maintained by additions of distilled water as required. After 60 minutes, the test panel shall be removed from the boiling solution, rinsed under flowing tap water (cold), rinsed in distilled water (20° - 26°C.), dipped in absolute ethyl alcohol, wiped dry with paper toweling, and reweighed. The presence of corrosion products shall be noted. Tests shall be run in duplicate.

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4.4.2.2.2 *Galvanic corrosion test procedure.* The aluminum test panels shall be cut 1-1/2 inches by 3/4 inches in size from approximately 0.034 inch thick sheet conforming to QQ-A-355, T3 or T4 temper. Sharp edges shall be smoothed with No. 4/0 - 150 alundum polishing paper. Both faces are polished with the same abrasive, stroking in one direction only. The magnesium test panels shall be cut 3 inches by 3/4 inch in size from approximately 0.034-inch thick sheet conforming to QQ-M-44. The magnesium panels shall be polished as the aluminum. Two 1/16 inch holes are made in each panel, close to the narrow ends of the aluminum panels, and at one end and the center of the magnesium panels, so that they may be fastened together, as shown in figure 1.

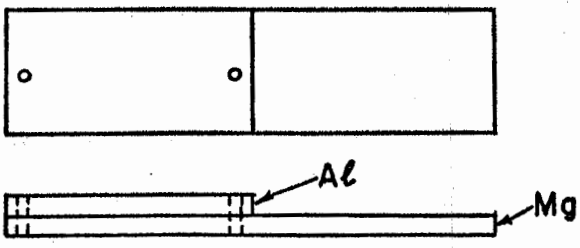


FIGURE 1. Galvanic corrosion test panels.

The panels shall then be cleaned as specified in 4.4.2.2, and weighed to the nearest 0.0001 gram. A magnesium and an aluminum panel are then fastened together by means of cotton string (figure 1), and immersed completely in 200 ml. of a boiling 4.75 percent (4.75 grams of compound per 100 ml. solution) distilled water solution of the compound contained in a 500-ml. tall-form beaker. The solution is kept at a boil for 2 hours, maintaining the solution level by additions of distilled water as required. The couple is then removed from the boiling solution, rinsed under flowing tap water (cold), unfastened, rinsed in distilled water, dipped in absolute ethyl alcohol, wiped dry with paper toweling, and the members of the couple reweighed. Tests shall be run in duplicate.

4.4.3 *pH value.*

4.4.3.1 *Apparatus.* The test shall be made with a pH electrometer having a sensitivity and readability of at least 0.05 pH. A sealed type, alkali-resistant glass electrode shall be used with a calomel reference electrode. The meter shall be standardized against a pH 10 buffer solution immediately before making a test. The instrument to be used shall be the Beckman pH meter laboratory model H, or equal, and the electrode shall be the Beckman "Type E" high pH glass electrode or equal.

4.4.3.2 *Procedure.* Prepare 100 ml. of a 4.75 percent (4.75 grams of compund in 100 ml. of solution) distilled water solution of the compound. Determine pH at 25°C. No correction for sodium ion concentration shall be made.

4.4.4 *Dust forming.* The inside walls of a 250-ml. glass stoppered graduate shall be rendered completely free of any grease and moisture. A 25-gram sample of the cleaner shall then be placed on the bottom of the clean tube. The tube shall then be stoppered, inverted, and immediately returned to its original position. After any suspended dust has been allowed to settle for 5 seconds, the stopper of the tube shall be removed and a moistened piece of red litmus paper suspended in the tube without touching the walls so that the lower end of the litmus paper strip coincides with the 210 ml. mark at the upper end of the graduate. After 60 seconds immersion, the litmus paper shall be removed and examined for any color change.

4.4.5 *Surface tension.* Dissolve 6.333 grams of the well-mixed compound in distilled water and dilute to volume in a 2-liter volumetric flask. Take a 10-ml. aliquot of this solution and dilute to volume in a 100-ml. volumetric flask. Determine the surface tension at 25°C., using a DuNouy tensiometer or interfacial tensiometer with a 6-cm. circumference platinum ring. The instru-

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ment shall be calibrated with weights, and Harkins-Jordon Correction Factors shall be applied. The solution shall stand 1 hour in test vessel before test is run.

4.4.6 *Penetration.* Place 100 grams of the compound in a flat bottom straight side, Pyrex glass crystallizing dish, 40 mm. high by 80 mm. diameter. Place sufficient lead shot in a 250-ml., low-form beaker so that the combined weight of the beaker and shot equals 800 grams. Place the weighted 250-ml. beaker on top of the 100 grams of compound, then place the assembly in a desiccator containing a saturated solution of potassium sulphate (K_2SO_4). Place the desiccator in an oven maintained at 43° to 46° C. for 24 hours. Remove assembly from oven and desiccator; remove the weighted 250-ml. beaker from compound, and allow crystallizing dish containing compound to cool at room temperature for 2 hours. Determine penetration at three points on the surface, each at least 12 mm. from the side of the crystallizing dish, using the penetrometer and grease penetrometer cone prescribed for method 311, Fed. Test Method Std. No. 791, with a 5-second hold, 250 grams total weight of the cone and attachments, and temperature of the compound at 24° to 25° C. Run test in duplicate and take average.

4.4.7 *Segregation.* Samples taken from different portions of the container shall be analyzed for one constituent of the compound, such as silicate by the appropriate method of 4.4.8.

4.4.8 *Composition.*

4.4.8.1 *Silicate.* Accurately weigh a 2 ± 0.001-gram sample and place in a 400-ml. beaker. Dry for 1 hour at 105°C. Cool to room temperature. Add 200 ml. of neutral, absolute ethyl alcohol, and digest on a steam bath, with frequent stirring, for 1 hour. Filter thru a Gooch crucible (asbestos mat), transferring and washing with neutral, absolute ethyl alcohol. Draw air thru crucible until odor of alcohol has disappeared. Place Gooch crucible in original 400-ml. beaker, add 150 ml. distilled water, bring to boil, and boil 10 minutes. Filter hot through a No. 4 Whatman paper into a 600-ml. beaker, washing with hot water. Acidify with concentrated HNO_3 (1.42 s.g.), and add 15 ml. of concentrated HNO_3 in excess. Digest on a hot plate for 15 minutes. Cool to room temperature. Add 15 ml. of $HClO_4$ (60%). Evaporate to fumes, and fume 15 minutes. Cool. Add 100 ml. distilled water plus 5 ml. concentrated HNO_3. Heat to dissolve all soluble matter. Filter hot through a No. 41 Whatman paper, washing free from acid with hot water. Transfer filter paper and residue to a platinum crucible. Dry at 105°C. Then ignite at 1000°C. Cool. Weigh. The percentage of silicate as SiO_2 is calculated as follows:

Percent silicate as SiO_2 = weight of residue × 50

4.4.8.2 *Total phosphates.*

4.4.8.2.1 *Preparation of molybdate solution.* Prepare the molybdate precipitating solution as follows:

No. X—Dissolve 100 g. c.p. ammonium molybdate in 400 ml. water plus 80 ml. concentrated NH_4OH.

No. Y—Dilute 400 ml. concentrated HNO_3 with 600 ml. water.

Pour No. X into No. Y slowly, with stirring, to form the molybdate solution (see 4.4.8.2.2).

4.4.8.2.2 *Test procedure.* Accurately weigh a 2- ± 0.001-gram sample and place in a 400-ml. beaker. Dry for 1 hour at 105°C. Cool to room temperature. Add 200 ml. of neutral, absolute ethyl alcohol, and digest on a steam bath, with frequent stirring, for 1 hour. Filter through a Gooch crucible, transferring with neutral, absolute ethyl alcohol. Draw air through crucible until odor of alcohol has disappeared. Finish drying the crucible and contents in 105°C. oven for 15 minutes. Place the Gooch crucible in the original 400-ml. beaker, add 150 ml. of water, bring to a boil and boil 10 minutes. Filter hot through

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a No. 4 Whatman paper into a 600-ml. beaker, washing with hot water. Acidify with concentrated HNO_3 (1.42 specific gravity), add 15 ml. of concentrated HNO_3 in excess. Digest on a low heat hot plate for 15 minutes. Cool to room temperature. Add 15 ml. of $HClO_4$ (60 percent). Evaporate to fumes, and fume 15 minutes. Cool. Add 100 ml. of water plus 5 ml. of concentrated HNO_3. Heat to dissolve all soluble matter. Filter through a No. 41 Whatman paper into a 500-ml. volumetric flask, washing with hot water until free of acid. Filtrate is cooled and diluted to volume. Place a 50-ml. aliquot in a 400-ml. beaker. Add 65 ml. of water, 15 g. of NH_4NO_3, and 10 ml. of concentrated HNO_3 in order. Heat to 60° to 70°C. Add slowly with stirring 70 ml. of molybdate solution (see 4.4.8.2.1). Let stand 1 hour. Check for completeness of precipitation. Filter through a tared Gooch crucible, using 1 percent NH_4NO_3 solution for transferring and washing free from acid. Precipitate is given a final wash with 3 ml. water. Dry at 105° C. for 2 hours. Cool and weigh.

Calculation:

$$\text{Percent phosphate as } P_2O_5 = \text{weight of precipitate} \times 18.9$$

4.4.8.3 *Total alkalinity.* Transfer a 10-gram sample to a 1-liter volumetric flask. Add about 350-ml. hot distilled water (88°C). Agitate to dissolve. Cool to room temperature. Dilute to 1-liter with distilled water. Pipette a 100-ml. aliquot into a 300-ml. Erlenmeyer flask. Add 3 drops of methyl orange indicator solution, and titrate with 0.5 *N* HCl. Total alkalinity as Na_2O is calculated as follows:

$$\text{Percent total alkalinity as } Na_2O = \text{ml. of HCl} \times 1.55$$

4.4.8.4 *Alcohol soluble matter.* Place a 10-gram sample in a 400-ml. beaker. Add 200 ml. neutral, absolute ethyl alcohol, and digest on a steam bath with frequent stirring for 30 minutes. Filter thru a Gooch crucible (asbestos mat), using suction, and retaining as much as possible of the residue in the beaker. Repeat the extraction-decantation procedure three times, using 25 ml. hot absolute ethyl alcohol each time. Evaporate the total alcohol filtrate to smaller volume, about 100-150 ml. Dry the residue in the beaker until free from alcohol odor. Add 10 ml. distilled water, and dissolve residue. Add 100 ml. absolute ethyl alcohol slowly and with stirring. Heat to boiling on steam bath, and filter thru original Gooch crucible into tared 250-ml. beaker holding the original alcohol filtrate (evaporated to smaller volume). Wash with hot absolute ethyl alcohol. Evaporate combined alcohol filtrate and washings on steam bath. Dry for 45 minutes at 80°C. ± 2°C. in a vacuum oven. Cool in desiccator and weigh as alcohol soluble matter. The calculation is:

$$\text{Percent alcohol soluble matter} = \text{weight alcohol soluble matter} \times 10$$

5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packing.

5.1.1 *Level A or B.* Cleaning compound shall be furnished in 5-gallon steel pails conforming to PPP-P-704, type II, class 3, or in 55-gallon steel drums conforming to PPP-D-729, type III or type IV, as specified in the contract or order.

5.1.2 *Level C.* Cleaning compound shall be furnished in accordance with standard commercial practice.

5.2 Marking.

5.2.1 *Military agencies.* Containers shall be marked in accordance with the requirements of MIL-STD-129.

5.2.2 *Civil agencies.* Containers shall be marked in accordance with Fed. Std. No. 123.

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5.2.3 *Additional marking.* The labeling shall comply with the requirements of the Federal Hazardous Substances Labeling Act. In addition to the marking specified above the following markings shall be included on each container:

> Directions for use—Use compound at a concentration of 7 ounce per gallon of water at a rolling boil for soak cleaning. For hydrosteam cleaning, dissolve compound completely in hot water to obtain a concentration of 7 ounces per gallon of water, and adjust concentrate solution metering device to obtain desired results. Hot water should be used for dissolving and solution of the compound must be complete before using. There must be no undissolved material in concentrate tank.

6. NOTES

6.1 **Intended use.** Cleaning compound covered by this specification is intended for use in the hot soak tank cleaning of ferrous and nonferrous alloy parts. Aluminum-magnesium assemblies may be cleaned with freedom from galvanic corrosion of the aluminum. Magnesium, when coupled with aluminum, may acquire a light brown coating, which, if objectionable, can be removed by a few seconds immersion in 5 percent nitric acid followed by water rinsing.

The compound may also be used in injection hydrosteam units using existing plant steam supplies for steam cleaning ferrous and nonferrous alloys.

The compound must not be used in steam cleaning machines of the high pressure-continuous tubular coil, vapor generator type.

6.1.1 *Improved cleaner (soak or hydrosteam).* The effectiveness of the cleaner can be increased by conversion of the aqueous solution of the cleaner into a stable, single phase, alkaline water-base organic solvent solution, which is operated at a boil when used as a soak detergent, or may be used as a conventional hydrosteam cleaner. The change is accomplished by adding 1 gallon of hexylene glycol (boiling point 192°-200°C., and specific gravity at 20°/20°C., 0.922-0.925) to 9 gallons of a 7 ounce per gallon water solution of the P-C-436b cleaner. The hexylene glycol content is maintained by periodic test, as follows:

To 50 ml. of the solution containing hexylene glycol contained in a 100-ml. graduated cylinder, add an excess of anhydrous potassium carbonate, analytical reagent grade. The hexylene glycol collects as the top liquid layer. The ml. in the top layer multiplied by two gives the percent by volume of hexylene glycol, which should be maintained at 9.0 to 11.0 percent.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Quantity of cleaning compound in pounds.
(*c*) Type, class, style, and capacity of containers.
(*d*) The level of packing required.
(*e*) Instructions regarding availability of inspection records (see 4.1).
(*f*) Type III or IV of 55-gallon drums, if applicable.

6.3 **Treatment of asphalt prior to soiling.** Bidders are cautioned against repeated heating of their entire stock of asphalt prior to spreading on the test panel. It is suggested that only enough asphalt be removed from the container on a spatula to soil one panel; soiling may be facilitated by slight heating (not over 50°C.) of both asphalt and panel.

6.4 **Composition.** The cleaning compound is contemplated to be comparable in performance to the following approximate

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composition by weight (see table I). However, the Government assumes no responsibility for the acceptance of a product claimed to be manufactured under the identical formula. Bidders are cautioned that their product will not meet the asphalt cleaning requirement unless the combination of surfactants is used; either surfactant alone will not impart the required detergency to the product.

TABLE I.—*Composition of cleaning compound*

Ingredient	Material specification	Percent by weight
Sodium metasilicate anhydrous	Fed. Spec. O-S-604, Type II	31.3
Primary sodium phosphate, monobasic, anhydrous	Technical grade NaH_2PO_4	12.3
Trisodium phosphate, anhydrous	Fed. Spec. O-S-642, Type I	24.8
Nonionic surfactant	MIL-D-16791, Type I	7.9
Anionic surfactant	Sodium alkyl benzene sulphonate (C_{12} to C_{18} alkyl), 40 percent active	23.7

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cleaning compounds, not otherwise indexed by name, granular.
Carload minimum weight 36,000 pounds.

Motor:

Cleaning compound, not otherwise indexed, other than liquid.
Truckload minimum weight 36,000 pounds, subject to Rule 115 National Motor Freight Classification.

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